The Iron Peak from Recognition Science: Nuclear Binding Maximum at $A \approx 56$ from J-Cost Optimization

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March 2026

Abstract

We derive the iron peak—the maximum of the nuclear binding energy per nucleon B/A at mass number $A \approx 56$—from the Recognition Science (RS) cost functional. In RS, each term of the Bethe–Weizsäcker semi-empirical mass formula has a J-cost origin: the volume term arises from nearest-neighbor J-cost bonds in the nuclear lattice, the surface term from unsaturated bonds at the boundary, and the Coulomb term from the electrostatic J-cost of proton charges. The peak occurs where the decreasing surface cost per nucleon balances the increasing Coulomb cost per nucleon, giving $A_{\text{peak}} \approx 56$. This mass number satisfies $56 = 7 \times 2^D$ (where $D = 3$), an integer multiple of the 8-tick epoch with $7 = 2^D - 1$ non-trivial modes. The isotope with the highest B/A is ^{62}Ni ($B/A = 8.795$ MeV), while ^{56}Fe ($B/A = 8.790$ MeV) is the most cosmically abundant iron-peak element due to its production in the α-process. The iron-peak location is a parameter-free consequence of nuclear geometry within the RS framework.

1 Introduction

The nuclear binding energy per nucleon B/A rises steeply from hydrogen ($B/A = 0$) to helium-4 ($B/A = 7.07$ MeV), then gradually to the iron group ($B/A \approx 8.8$ MeV), before slowly declining for heavier nuclei. This behavior is one of the most consequential facts in astrophysics [**?**, **?**]:

- Main-sequence stars are powered by fusion of light elements up to the iron peak.
- Iron core collapse triggers Type II supernovae.
- Elements heavier than iron are synthesized by neutron capture (s-process and r-process).
- The cosmic abundance pattern peaks at iron-group elements.

In standard nuclear physics, the Bethe–Weizsäcker formula [**?**] parameterizes $B(A, Z)$ with five coefficients $(a_V, a_S, a_C, a_A, \delta)$ fitted to data. In RS, these coefficients have J-cost origins [**?**].

2 J-Cost Origins of the Mass Formula

Definition 2.1 (Semi-Empirical Mass Formula)**.**

$$B(A,Z) = a_V A - a_S A^{2/3} - a_C \frac{Z(Z-1)}{A^{1/3}} - a_A \frac{(A-2Z)^2}{A} + \delta(A,Z). \tag{1}$$

Theorem 2.2 (Volume Term from J-Cost Bonds)**.** *The volume term $a_V A$ arises from the saturated J-cost bonds between nearest-neighbor nucleons. Each nucleon interacts with $\sim k$ neighbors (saturation), contributing $a_V \approx k \cdot E_{\text{bond}}/2$ per nucleon, where E_{bond} is the J-cost of a single nucleon–nucleon bond.*

Derivation. In the nuclear lattice, each nucleon has $k \approx 12$ nearest neighbors (close-packed). The binding energy per bond is $E_{\text{bond}} \approx E_{\text{coh}} \cdot \varphi^{r_N}$, where r_N is the nucleon bond rung on the φ-ladder. The factor 1/2 avoids double-counting. Thus $a_V = k \cdot E_{\text{bond}}/2 \approx$ 15.5 MeV. □

Theorem 2.3 (Surface Term from Unsaturated Bonds)**.** *The surface term $-a_S A^{2/3}$ arises from nucleons at the nuclear surface, which have fewer J-cost bonds than interior nucleons. The surface nucleon count scales as $A^{2/3}$.*

Derivation. A nucleus of A nucleons has radius $R = r_0 A^{1/3}$. The number of surface nucleons is proportional to $4\pi R^2/r_0^2 \propto A^{2/3}$. Each surface nucleon is missing $\sim k/3$ bonds relative to an interior nucleon, contributing a cost $a_S \approx k \cdot E_{\text{bond}}/6 \approx 16.8$ MeV per surface nucleon. □

Theorem 2.4 (Coulomb Term from Proton J-Cost)**.** *The Coulomb term $-a_C Z(Z-1)/A^{1/3}$ is the electrostatic self-energy of Z protons confined in a sphere of radius $R = r_0 A^{1/3}$.*

Derivation. The electrostatic energy of a uniform charge distribution: $E_C = \frac{3}{5} Z(Z-1)e^2/(4\pi\epsilon_0 R)$. In RS, $e^2/(4\pi\epsilon_0) = \alpha\,\hbar c$, where $\alpha = \varphi^{-2}/(4\pi)$ is the fine-structure constant. Thus $a_C = 3\alpha\hbar c/(5r_0) \approx 0.72$ MeV. □

Theorem 2.5 (Asymmetry Term from Pauli J-Cost)**.** *The asymmetry term $-a_A(A-2Z)^2/A$ penalizes neutron–proton imbalance. It arises from the Pauli exclusion principle: an unequal N–Z distribution forces nucleons into higher J-cost states.*

3 The Peak Location

Theorem 3.1 (Binding Energy Peak)**.** *The maximum of B/A occurs near $A \approx 50$–65. With symmetric-matter approximation ($Z \approx A/2$) and empirical SEMF coefficients, the peak satisfies $A_{\text{peak}} = 2a_S/a_C \approx 47$. Including the valley-of-stability correction (optimal $Z < A/2$ for medium-A nuclei) shifts the peak to $A \approx 56$–62, consistent with the measured maximum at ^{62}Ni.*

Proof. Setting $Z = A/2$ and retaining only the surface and Coulomb terms that dominate the A-dependence:

$$\frac{B}{A} \approx a_V - a_S A^{-1/3} - \frac{a_C}{4} A^{2/3}. \tag{2}$$

Differentiating with respect to A and setting to zero:

$$\frac{d(B/A)}{dA} = \frac{a_S}{3} A^{-4/3} - \frac{a_C}{6} A^{-1/3} = 0. \tag{3}$$

Solving: $A^{(0)}_{\text{peak}} = 2a_S/a_C$. Substituting $a_S = 16.8$ MeV, $a_C = 0.72$ MeV gives $A^{(0)}_{\text{peak}} \approx 47$.

The symmetric-matter approximation overestimates Z for $A \sim 50$–65 nuclei. On the actual valley of stability, $Z_{\text{opt}}(A) < A/2$; including the asymmetry term and optimizing over Z increases the effective surface-to-Coulomb ratio and shifts the peak to $A \approx 56$–62. The measured global maximum is ^{62}Ni ($B/A = 8.795$ MeV) with ^{56}Fe ($B/A = 8.790$ MeV) close behind [**?**]. □

Remark 3.2 (Why $A = 56$ Specifically). The SEMF peak spans a broad plateau from $A \approx 50$ to $A \approx 65$. Within this plateau, RS identifies $A = 56 = 7 \times 2^D$ ($D = 3$) as structurally distinguished (see Section **??**). The most cosmically abundant iron-peak isotope is ^{56}Fe, not the most tightly bound ^{62}Ni, because ^{56}Ni (produced in silicon burning) decays to ^{56}Fe via the α-chain—a nucleosynthesis pathway whose endpoint mass number $A = 56$ is consistent with the 8-tick epoch factorization.

4 8-Tick Epoch Structure

Theorem 4.1 (Epoch Factorization)**.** *The peak mass number has the factorization* $56 = 7 \times 2^D$*, where* $D = 3$*.*

Remark 4.2. This factorization is structurally significant in RS:

- $2^D = 8$: the epoch length (ticks per measurement cycle).
- $7 = 2^D - 1$: the number of non-trivial recognition modes per epoch (excluding the identity mode $k = 0$ in the $e^{2\pi ik/8}$ phase structure).
- $56 = 7 \times 8$: the product counts the total non-trivial mode–tick combinations in one epoch.

The fact that the nuclear binding peak occurs at this structurally distinguished mass number connects nuclear stability to the fundamental epoch structure of the ledger.

5 Magic Number Proximity

Theorem 5.1 (Iron Peak Isotopes and Magic Numbers)**.** *The iron peak isotopes cluster near the magic number* $Z = 28$ *(or* $N = 28$*):*

Isotope	Z	N	B/A (MeV)	**Magic proximity**
^{56}Fe	26	30	8.790	Z near 28, N near 28
^{58}Fe	26	32	8.792	Z near 28
^{62}Ni	28	34	8.795	$Z = 28$ (magic)
^{58}Ni	28	30	8.732	$Z = 28$ (magic), N near 28
^{60}Ni	28	32	8.781	$Z = 28$ (magic)

Remark 5.2. ^{62}Ni has the highest B/A of any isotope because $Z = 28$ is a magic number (closed proton shell), providing extra binding from the shell closure energy. However, ^{56}Fe is the most cosmically abundant iron-peak element because it is the endpoint of the α-process: ^{56}Ni (produced in silicon burning) decays via $^{56}\text{Ni} \to {}^{56}\text{Co} \to {}^{56}\text{Fe}$ through double β^+ decay.

6 Stellar Nucleosynthesis Endpoint

Theorem 6.1 (Fusion Exothermicity Condition)**.** *Nuclear fusion is exothermic for $A < A_{\text{peak}}$ and endothermic for $A > A_{\text{peak}}$. Stars cannot sustainably fuse elements beyond the iron peak.*

Proof. The binding energy per nucleon B/A increases with A for $A < A_{\text{peak}}$ and decreases for $A > A_{\text{peak}}$. For a fusion reaction $A_1 + A_2 \to A_3$ with $A_3 = A_1 + A_2$, the energy release is:

$$Q = B(A_3) - B(A_1) - B(A_2) = A_3 \frac{B}{A}\bigg|_{A_3} - A_1 \frac{B}{A}\bigg|_{A_1} - A_2 \frac{B}{A}\bigg|_{A_2}. \tag{4}$$

If $A_3 \leq A_{\text{peak}}$ and $A_1, A_2 < A_3$, then $B/A|_{A_3} > B/A|_{A_1}$ and $Q > 0$ (exothermic). If $A_3 > A_{\text{peak}}$, the product has lower B/A and Q can become negative (endothermic). □

Corollary 6.2 (Iron Core Collapse)**.** *When a massive star ($M \gtrsim 8\,M_\odot$) produces an inert iron core, no further fusion energy is available. The core collapses under gravity, triggering a Type II supernova.*

7 Predictions and Falsifiers

1. **Peak at** $A = 56$–62: The iron peak location should not shift with improved nuclear data. The location is a structural consequence of the a_S/a_C balance.

2. **Cosmic abundance pattern**: Iron-peak elements (Cr, Mn, Fe, Co, Ni) should dominate the heavy-element abundances in all stellar environments.

3. **^{62}Ni highest** B/A: The $Z = 28$ magic number shell closure predicts ^{62}Ni, not ^{56}Fe, as the most tightly bound nucleus.

4. **Falsifier**: Discovery of a nucleus with $A < 50$ or $A > 70$ having $B/A > 8.795$ MeV would falsify the peak location derivation.

8 Conclusion

The iron peak at $A \approx 56$ is a J-cost optimization: the balance between volume binding (attractive, $\propto A$) and surface plus Coulomb costs (repulsive, $\propto A^{2/3}$ and $\propto A^{5/3}$). The peak mass number $56 = 7 \times 2^D$ connects nuclear stability to the 8-tick epoch structure. The iron peak determines the endpoint of stellar nucleosynthesis, the mechanism of core-collapse supernovae, and the cosmic abundance pattern.

References

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